Connectome-Constrained Spiking Networks Reproduce Emergent Computations in Larval Olfactory Pathway

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Abstract

The complete synaptic connectome of the larval *Drosophila melanogaster* brain offers an opportunity to build structurally faithful neural circuit models. However, the connectome provides only the wiring diagram, not the biophysical dynamics or functional logic underlying neural computation; one approach to bridging this gap is supervised training on a biologically relevant task, with the resulting neural activity evaluated against existing data. We present a spiking neural network model of the larval olfactory pathway, from odor receptors (ORs) via the antennal lobe to Kenyon cell (KC) activation in the mushroom body (MB). All connectivity is fixed by the connectome, complemented by biologically plausible neuron dynamics including spiking and non-spiking neurons, two-compartment KCs, gap junctions, synaptic depression, and axo-axonic connections. Despite this richness, the model requires only 449 free biological parameters, which are trained on a 28-odor classification task using electrophysiological OR response data as input. We use two-phase artificial-to-spiking neural network transfer learning: a rate-based teacher is trained to convergence, then its weights are transferred to the spiking network and fine-tuned with surrogate gradient descent under full biophysical constraints for accuracy and realistic KC activation sparsity. The trained model spontaneously reproduces decorrelation localized to the MB, anterior paired lateral neuron suppression of KC activity by $\sim$50%, concentration-invariant gain control across a 17-fold concentration range, and sub-additive KC responses to odor mixtures matching experimental observations. An ensemble of independently trained models converges to consistent KC representations and parameters, demonstrating that the connectome dominates the learned solutions. Additionally, systematically perturbing circuit components reveals their individual contributions and generates testable hypotheses. We demonstrate that the connectome, combined with biophysical constraints and a simple learning objective, produces circuit-level computations matching independent biological measurements. This establishes a generalizable methodology beyond larval olfaction for translating connectome data into testable spiking models to determine microcircuit functional logic and the roles of constituent neurons.[1]

[1] Code and data availability: https://github.com/InsectRobotics/connectome_emergence.git

Introduction

Recent advances in electron microscopy (EM) have delivered a revolution in connectomics. The annotation of 3,016 neurons and 548,000 synapses in the complete EM reconstruction of the larval *Drosophila melanogaster* brain (Winding et al., 2023) was quickly followed by the full adult *Drosophila* brain reconstruction describing 5×10^7 chemical synapses between 139,255 neurons (Dorkenwald et al., 2024; Schlegel et al., 2024), representing the most comprehensive wiring diagram of any animal brain to date. However, a wiring diagram is not a circuit model. These connectomes do not directly reveal the computations a circuit performs, the dynamics it produces, or the behaviors it generates.

There are two clear challenges to developing a methodology that maximally exploits the latent information in this data. First, connectomes provide connectivity but lack biophysical details, such as membrane time constants, spike thresholds, synaptic strengths in physiological units, gap junction conductances, or short-term plasticity parameters. These must be inferred, estimated, or learned as part of any methodology developing connectome data into a computational model. Second, the scale and complexity of connectome-constrained models demand biologically informed training and validation strategies.

One connectome-constrained, task-optimized approach used to model the adult *Drosophila* visual system involves a deep mechanistic network where the connectivity between 64 cell types in the optic lobe motion pathways was fixed to experimentally measured values, while unknown single-neuron and synapse-type parameters were optimized via deep learning to perform optic flow estimation (Lappalainen et al., 2024). Ensembles of task-optimized deep mechanistic networks predicted neural tuning properties that agreed with experimental measurements across 26 independent studies, without explicitly imposing these properties during training. This indicates that connectome structure, combined with a known computational objective and gradient-based optimization, can bridge the gap between static wiring and neural dynamics. Subsequent connectome-constrained models include finer biological detail, such as gap junctions (Chung & Kim, 2025), the raw connectome without

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simplifications (Duan et al., 2026), or spiking neurons, but without task-based training to allow for emergent computations (Wang et al., 2026).

We combine all these principles into a spiking model of the larval *Drosophila* olfactory system with task-based training. This is a compact sensory circuit that maps 21 odor receptor (OR) types through several processing stages in the antennal lobe (AL) and mushroom body (MB) to produce sparse, decorrelated odor representations suitable for associative learning (Lin et al., 2014). The circuit is small but biophysically rich, incorporating gap junctions, lateral excitatory connections, and diverse local interneuron subtypes not considered in Lappalainen et al. (2024). Even with these additional features and compartment-resolved Kenyon cell (KC) connectivity, our model has only 449 free biological parameters.

Using end-to-end training on a 28-odor classification task, with a rate-based teacher and surrogate gradient descent for fine-tuning, we show multiple emergent circuit properties that can be compared to independent biological benchmarks (Asahina et al., 2009; Bhandawat et al., 2007; Honegger et al., 2011; Mancini et al., 2023).

The systematic perturbation of individual circuit components reveals their functional contributions, generating experimentally testable hypotheses directly from the trained model. This all suggests that connectome-constrained optimization is a viable methodology to extract functional principles of neural computation from anatomical data.

Methods

Overview

Can connectome data, combined with biophysical constraints and a simple learning objective, give rise to circuit-level computational models that match independent biological measurements? To answer this question, we built a spiking model of the larval *Drosophila* olfactory pathway with connectivity fixed by the Winding et al. (2023) connectome and trained for odor identification (Fig. 1). The model receives a 21-dimensional recorded OR response vector (Kreher et al., 2008) and predicts which of 28 odorants produced it. The OR vector is injected as a constant current into 42 olfactory receptor neurons (ORNs; 21 types $\times$ 2 hemispheres), which drive the spiking circuit for 30 simulation timesteps (30 ms, with Δt = 1 ms) as the input is processed through the AL and into the MB KCs. A linear decoder reads out the time-averaged KC spike rates to produce a 28-way classification. The model is trained using a two-phase ANN-to-SNN (artificial-to-spiking neural network) transfer learning approach, repeated independently for an ensemble of five models to test whether the connectome constrains the learned solution. Decorrelation, calibrated anterior paired lateral (APL) neuron inhibition, concentration invariance, and odor mixture coding are evaluated post-hoc against independent biological benchmarks not included in the loss function, and the contribution of specific circuit features is evaluated by perturbation experiments.

Biological Data

Connectome: Neurons and synaptic connectivity were extracted from the complete larval *Drosophila* connectome (Winding et al., 2023). We identified all neurons in the olfactory pathway across both hemispheres by cell-type annotation: 42 ORNs, 108 local neurons (LNs), 72 projection neurons (PNs), 144 Kenyon cells (KCs), and 2 APL interneurons. These 368 neurons are connected by a total of $\sim$50,000 synapse contacts across $\sim$12,000 synaptic edges, providing the architecture for our model (Fig. 1). Only a single parameter is learned for each pre- and postsynaptic neuron type pair under a unitary synapse assumption.

Odorant Responses: ORN response profiles were taken from Kreher et al. (2008), which provides electrophysiological recordings of 21 larval OR types responding to 28 odorants (Table S1). Responses were normalized to [0, 1] and used as input to our model. Multiplicative OR noise of 30% coefficient of variation (CV) was used to generate data from these OR response profiles. For each training epoch, 10 noisy repetitions of each odor are generated (280 samples per epoch); test performance is evaluated on 50 independent noise draws per odor (1400 samples).

Spiking Neuron Model Features

LIF neurons: We used leaky integrate-and-fire (LIF) neurons with exact exponential integration, learnable per-neuron spike thresholds (distinct connectivity patterns warrant distinct excitability), a shared membrane time constant per population (within-type similarity) (Gouwens & Wilson, 2009), and fixed V_{reset} and τ_{ref} (little biological variability). The choice of which parameters to learn follows Lappalainen et al. (2024), with spiking neuron-specific parameters from standard LIF parameterizations (Jürgensen et al., 2021):

$$V_\infty = V_{\text{rest}} + R_m\, I_{\text{syn}}(t) \tag{1}$$

$$V_i(t+\Delta t) = V_\infty + \left(V_i(t) - V_\infty\right)\exp\left(-\frac{\Delta t}{\tau_m}\right) \tag{2}$$

$$S_i(t) = \begin{cases} 1 & \text{if } V_i(t) \geq V_{\text{th},i} \text{ and } t - t_i^{\text{last}} > \tau_{\text{ref}} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

$$V_i(t) \leftarrow V_{\text{reset}} \quad \text{if } S_i(t) = 1 \tag{4}$$

Eq. 1 defines the steady-state membrane voltage, V_∞, from total synaptic input current, $I_{\text{syn}}(t)$; Eq. 2 updates the membrane voltage via exponential decay toward V_∞; Eq. 3 is the spike function, which fires when voltage exceeds threshold outside the refractory period; and Eq. 4 resets the voltage after a spike.

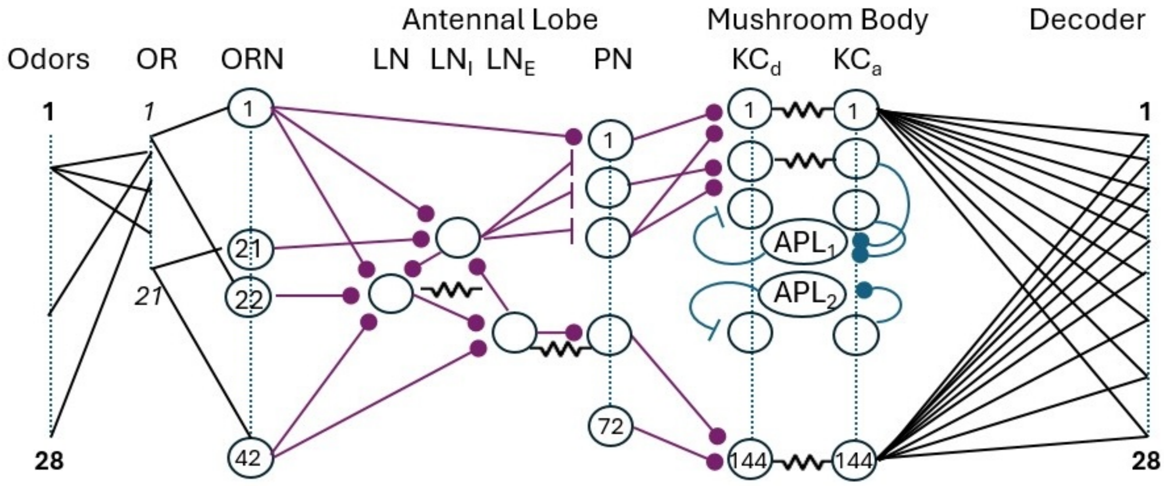

Figure 1: Schematic of the olfactory circuit. 28 input odors are encoded as patterns across 21 odor receptor (OR) types. These activate (one-to-one in each hemisphere) olfactory receptor neurons (ORNs) projecting to the antennal lobe (AL). The AL includes local neurons (LNs) and projection neurons (PNs). LNs contact each other with chemical and electrical synapses (gap junctions) and some contact PNs either with 'broad' inhibition (LN_i) or 'picky' excitation (LN_e), also with gap junctions. PNs project to Kenyon cells (KCs) in the mushroom body. KCs are modeled with separate dendrite KC_d and axon KC_a, with electrical coupling. Half the KCs connect to each non-spiking anterior paired lateral (APL) neuron, which provides inhibitory feedback. For training, the KCs are fully connected to a decoding layer of 28 outputs. Additional connections not shown include gap junctions between PNs that share ORN input and axo-axonic connections between KCs. Note that all neural nodes (circles/ovals) and their connections are defined by the connectome. See text and Table 1 for the learnable parameters.

$V_i(t)$ is the membrane voltage of neuron i; $V_{\text{rest}} = V_{\text{reset}} = -60$ mV; $R_m = 1$ GΩ; τ_m is the membrane time constant (learnable per population, bounded to 5–50 ms); $S_i(t) \in \{0, 1\}$ is the spike indicator (Eq. 3), equal to 1 when neuron i fires and 0 otherwise; $V_{\text{th},i}$ is the spike threshold (learnable per neuron, bounded to $[-55, -30]$ mV); and $\tau_{\text{ref}} = 2$ ms (fixed refractory period). We use a simulation timestep of $\Delta t = 1$ ms, which is standard for LIF simulations (Bellec et al., 2020) and matches the millisecond timescale of membrane dynamics and spike generation (Kistler & Gerstner, 2002). During training, the non-differentiable spike function is replaced by a sigmoid surrogate gradient to enable backpropagation (Neftci et al., 2019).

Graded APL interneuron: The APL neuron uses graded (non-spiking) transmission, integrating KC activity through a learnable time constant (τ_{APL}, bounded to 10–50 ms), and providing global divisive inhibitory feedback to all KCs through connectome-specified synaptic connections (Amin et al., 2020; Mancini et al., 2023). $I_{\text{KC},i}(t) = I_{\text{in},i}(t)/\left(1 + g_{\text{APL}}\, a_{\text{APL}}(t)\right)$, where $a_{\text{APL}}(t)$ is the APL activity and g_{APL} the learnable APL output gain.

Two-compartment KCs: KCs are modeled with separate dendritic (receives PN input) and axonal (spike-generating) compartments coupled by a learnable soma conductance (g_{soma}, bounded to 1–20 nS). This reflects the anatomical separation of the MB calyx (KC dendrites) and lobes (KC axons).

Short-Term Synaptic Depression: Tsodyks–Markram short-term depression (STD) was applied to all chemical synaptic connections (Tsodyks & Markram, 1997). Each synapse type has two learnable parameters: recovery time constant (τ_{rec}) and release probability (U). STD provides activity-dependent gain control: stronger presynaptic drive depletes vesicles faster, automatically compressing the dynamic range. The dynamics follow $x_j(t+\Delta t) = 1 - (1 - x_j(t)) \exp(-\Delta t/\tau_{\text{rec}})$ for vesicle recovery, $I_{\text{post},i}(t) = \sum_j W_{ij} S_j(t)\, U\, x_j(t)$ for synaptic transmission, and $x_j(t) \leftarrow x_j(t) - U\, x_j(t)\, S_j(t)$ for vesicle depletion, where $x_j \in [0, 1]$ is the available vesicle fraction, τ_{rec} is the recovery time constant (20 ms–2 s), U is release probability (0.05–0.8), and W_{ij} is the connectome-fixed synaptic weight with learnable scalar strength.

Excitatory and Inhibitory LN Subtypes: Berck et al. (2016) identified inhibitory Broad LNs (LN_i; GABAergic, high fan-out, widespread lateral inhibition) and excitatory Picky LNs (LN_e; glutamatergic, low fan-out, selective excitation) as the two subtypes of AL LNs. We classified

LNs by their fan-out in the Winding et al. (2023) connectome, and of 108 LNs, only 26 have LN→PN connections. Berck et al. (2016) reports a 5:7 (LN_e:LN_i) ratio; using the closest discrete fan-out threshold ($\leq$ 16 PN targets), 9 neurons were classified as LN_e, and the remaining 17 as LN_i. The LN→PN connectivity matrix was thus split into inhibitory and excitatory pathways, each with independently learned synaptic strengths. As this 9:17 ratio differs from the 5:7 ratio reported in (Berck et al., 2016), we also tested LN ratios of 12:14, 6:20, and 13:13. These differences had no effect on any of the conclusions described below and left the results unchanged except for a slight increase in the MB localization of decorrelation for more excitatory ratios (13:13, 12:14).

Gap Junctions: Electrical synapses are a critical component of *Drosophila* AL circuitry, synchronizing membrane potentials between connected neurons (Huang et al., 2010; Kazama & Wilson, 2009; Yaksi & Wilson, 2010). Because the Winding et al. (2023) connectome does not include gap junction locations, we constructed proxy topologies for three documented gap-junction types based on known biology, and modeled their current as $I_{\text{gap}} = g \cdot (V_{\text{pre}} - V_{\text{post}})$, with one learnable conductance g per type.

Since LNs are electrically coupled (Fuenzalida-Uribe et al., 2025), we symmetrized the chemical LN→LN connectivity matrix to place gap junctions. LN_e couple electrically to their target PNs as a primary output mechanism (Yaksi & Wilson, 2010). PNs that share ORN input innervate the same glomerulus and are known to be electrically coupled (Kazama & Wilson, 2009). We identified sister PN–PN pairs by computing which PNs receive input from at least one common ORN. KC–KC gap junctions were excluded despite having been documented in adult *Drosophila* (Liu et al., 2016; Shyu et al., 2019), as direct evidence for KC–KC electrical coupling in the larval MB is currently lacking.

Biologically Realistic Noise: Six noise sources were applied during training to reflect the stochasticity inherent in biological neural circuits (Faisal et al., 2008): multiplicative OR noise (30% CV), membrane voltage fluctuations (1.0 mV standard deviation), background current (15 pA standard deviation), stochastic synaptic transmission (25% multiplicative CV on synaptic current), spike threshold jitter (1.0 mV standard deviation), and ORN input noise (10% CV).

Biological Parameter Clamping: All neuron parameters were clamped to biologically plausible ranges after each optimizer step: $V_{\text{th}} \in [-55, -30]$ mV, $\tau_m \in [5, 50]$ ms, $g_{\text{soma}} \in [1, 20]$ nS, $\tau_{\text{APL}} \in [10, 50]$ ms, $\tau_{\text{ref}} = 2$ ms (fixed). These are consistent with electrophysiological measurements in *Drosophila* olfactory neurons (Gouwens & Wilson, 2009). This prevents the optimizer from exploiting biologically implausible parameter regimes while allowing meaningful task-driven adaptation within realistic bounds.

Training Protocol

Training involved a two-phase transfer learning approach (Rathi et al., 2020), implemented in PyTorch (Paszke et al., 2019) with the Adam optimizer (Kingma & Ba, 2014), with downstream analyses performed using scikit-learn (Pedregosa et al., 2011).

Phase 1—Rate-based teacher: A non-spiking, rate-based model with identical connectome topology was trained using standard backpropagation on a 28-odor classification task (cross-entropy + sparsity loss). This training used the methodology developed in Lappalainen et al. (2024) and achieved ∼93% accuracy, providing a good initialization for the spiking student.

Phase 2—Spiking student: The 449 learnable biological parameters of the spiking model (Table 1) were instantiated from the teacher with some scaling. The spiking model was then fine-tuned with cross-entropy loss plus a differentiable sparsity loss: $\mathcal{L}_{\text{sparse}} = \left(\frac{1}{N}\sum_i \sigma((r_i - 0.02) \times 50) - 0.05\right)^2$, where r_i is the mean firing rate of KC i and σ is the sigmoid function. This penalizes deviations from 5% KC activation, serving as a differentiable proxy for metabolic energy constraints that limit biological firing rates but are absent from the model. Spikes are generated using a soft surrogate gradient function ($\sigma(x/v_s)$ with $v_s = 1$ mV) to enable backpropagation during training (Neftci et al., 2019). Zenke and Ganguli (2018) demonstrate similar performance across surrogate forms (sigmoid, exponential, piecewise-linear), supporting our use of the sigmoid. Gradient clipping and per-layer learning rate multipliers compensate for gradient attenuation through spiking layers. Each spiking model was trained from an independently trained teacher, ensuring that observed cross-model consistency reflects connectome constraints rather than shared initialization.

Results

Classification Accuracy and KC Sparsity

We trained connectome-constrained, biophysically inspired spiking models of the larval *Drosophila* olfactory pathway on the odor classification task. Training was applied independently to produce an ensemble of five spiking models which achieved 70.3% ± 1.1% test accuracy on 28-odor classification (∼20× random chance), with KC sparsity of 12.5% ± 0.1% (Fig. S1) compared to a biological target of 5–10% (Lin et al., 2014). Sparsity is the fraction of KCs emitting at least one spike during a 30 ms odor presentation.

Per-odor analysis reveals structured sparse coding: different odors activate distinct KC subsets, with ∼88% of

Table 1: Learnable Parameters

Parameter	Count
Input & membrane parameters	
OR gain (per receptor type)	21
Global OR$\rightarrow$ORN gain	1
ORN / LN / PN / APL τ_m (per population)	4
ORN v_{th} (per neuron)	42
LN v_{th} (per neuron)	108
PN v_{th} (per neuron)	72
KC v_{th} (per neuron)	144
AL chemical synapses (w, τ_{rec}, U each), axo-dendritic (AD) and non-axo-dendritic (non-AD)	
ORN$\rightarrow$LN / LN_i $\rightarrow$PN / LN_e $\rightarrow$PN / LN$\rightarrow$LN / PN$\rightarrow$LN / PN$\rightarrow$KC	$6 \times 3 \times 2$
ORN$\rightarrow$PN (AD only)	3
LN$\rightarrow$ORN (non-AD only)	3
KC & MB parameters	
KC$\rightarrow$KC axo-axonic and axo-dendritic synapses (w, τ_{rec}, U each)	3×2
Gap junction g: LN–LN, PN–PN, LN_e–PN / KC soma conductance g_{soma}	4
KC–KC dendro-dendritic / dendro-axonic pathway gain	2
KC$\rightarrow$APL strength / KC dendrite$\rightarrow$APL gain / APL output gain	3
Decoder	
Decoder weights (144×28)	4,032
Decoder bias	28
Circuit subtotal	**449**
Decoder subtotal	**4,060**
Total	**4,509**

KCs silent for any given stimulus. Sparsity correlates with OR response strength, so odors with strong, distinctive OR profiles produce denser KC activation.

A zero-parameter nearest-centroid classifier was used to test whether classification performance resides in the KC representations or depends on the trained linear decoder. For each odor, we computed the centroid as the mean KC spike pattern (a 144-dimensional vector) across all training samples. Test samples were then classified by cosine similarity, the standard metric for evaluating expansion coding in sparse population codes (Bernardi et al., 2020; Jürgensen et al., 2021; Srinivasan et al., 2023), i.e., each test KC pattern was assigned to the odor with the nearest training centroid. This classifier has no learnable parameters, instead exploiting the geometry of the KC representation space. The centroid classifier achieves 65.9% $\pm$ 0.5%, demonstrating that the KC representations themselves carry significant odor identity information, while confirming that the MB produces well-separated, linearly decodable odor representations.

Per-odor analysis reveals a variation in classification difficulty that reflects known properties of the input data, with several odors consistently misclassified for identifiable biological reasons (Fig. S2). Carbon dioxide (CO_2) achieves low accuracy because it is detected by the gustatory receptors Gr21a and Gr63a, not by any of the 21 ORs in our model (Jones et al., 2007; Kwon et al., 2007). The ester cluster is a group of odors whose OR response profiles we find to be nearly identical. Four odor pairs have pairwise cosine similarities exceeding 0.90, making them nearly indistinguishable at the receptor level. A real larva likely perceives these as a single "fruity ester" category rather than six distinct chemicals. Weak-response odors such as sfr (spontaneous firing rate, i.e., no odor), geranyl acetate, and propanoic acid have flat or low-amplitude OR profiles that blur with baseline noise; these same weakly encoded odors also yield the least reproducible KC representations across noisy presentations (Fig. S3). Despite these inherent biological constraints on discriminability, the model achieves $\sim$70% accuracy, which is quite reasonable given that larval chemotaxis assays yield moderate response indices even for strong attractants. Kreher et al. (2008) report response indices ranging from -0.40 (geranyl acetate, repellent) to 0.74 (E2-hexenal, strongest attractant), indicating that behavioral discrimination is imperfect even for the biological circuit.

Emergent Computations

Decorrelation is Localized to the Mushroom Body: A proposed function of the olfactory circuit, and more specifically of the divergent PN$\rightarrow$KC projections, is to produce sparse, high-dimensional KC representations in which overlapping odor-evoked patterns are decorrelated to support discriminative learning (Bhandawat et al., 2007; Caron et al., 2013; Lin et al., 2014). Our model reproduces this prediction quantitatively, with $\sim$4% decorrelation (based on cosine similarity ratios between processing stages) occurring in the AL, compared to $\sim$35% in the MB, with the structure of inter-odor cosine similarities preserved in KC spike activity (Fig. 2). We quantify decorrelation per odor pair as one minus the ratio of downstream to upstream cosine similarity (AL: PN-to-OR; MB: KC-to-PN).

APL Inhibition Matches Biological Measurements: To assess if our model could reproduce a quantitative benchmark for the inhibitory function of the non-spiking APL interneurons, we mimicked optogenetic APL activation as described in Mancini et al. (2023). In the baseline condition, we injected a weak excitatory current (1×10^{-10} A) into all 144 KCs to mimic the carbachol-induced spontaneous activity used in the biological experiment, while the APL received no additional drive. In the APL-activated condition, the same KC current was applied but the APL additionally received a proxy depolarizing current (0.7 in the model's internal APL activation units, chosen to stimulate the APL without saturat-

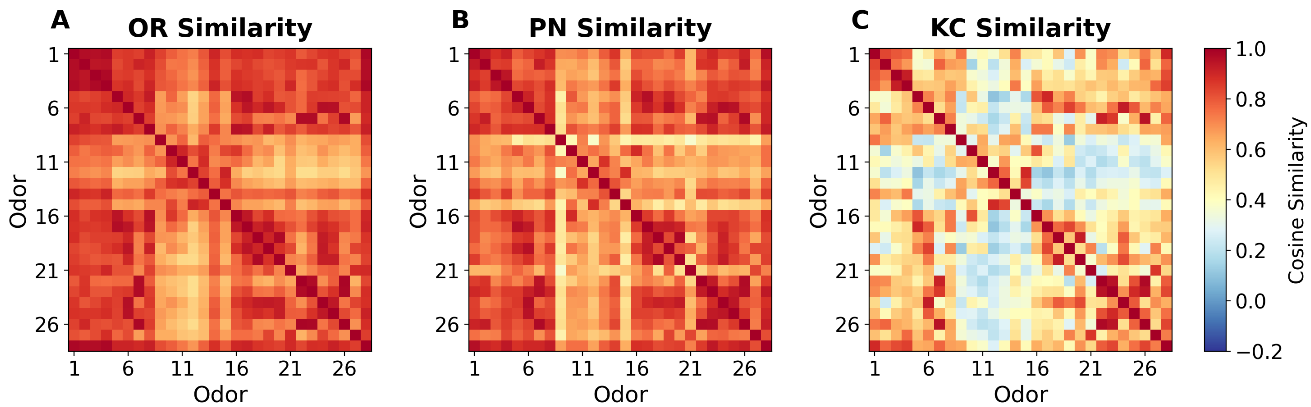

Figure 2: Progressive decorrelation of odor representations across the olfactory pathway (A–C) is shown using pairwise cosine similarity matrices for the 28 odors at successive processing layers, averaged over five independently trained models (20 noisy trials each). Odors 9–15 represent a distinct group of aromatic compounds, which manifests in the lack of similarity between them and the non-aromatic odors. (A) OR input shows high inter-odor similarity. (B) PN output after AL processing shows modest decorrelation ($\sim$4%). (C) KC output after MB expansion and APL inhibition shows substantial decorrelation ($\sim$35%), reducing total pairwise similarity by $\sim$39% relative to OR input.

ing it), simulating the optogenetic APL activation used by Mancini et al. (2023). We then measured total KC spike counts across 20 trials per condition and computed the ratio of baseline to APL-activated KC spike counts. Across the five-model ensemble, the KC spike ratio is 1.85 ± 0.05, corresponding to $\sim$46% KC suppression and closely matching the biological data of $\sim$50%, where the APL suppresses KCs meaningfully but not entirely.

Concentration Invariance: Another proposed function of this circuit is concentration invariant odor identification, achieved through gain control. We used a single concentration in training, and tested whether concentration invariance had emerged in the trained models. We tested the circuit's generalization for relative concentrations of c = 0.03–5.0, using Hill dose-response scaling on the input. Several biological predictions are confirmed by the model (Fig. 3). First, PNs exhibit sublinear gain, with their dynamic range (1.23$\times$) compressed below the OR input range (1.75$\times$), likely by LN inhibition and STD, consistent with Olsen et al. (2010). APL-mediated normalization compresses KC dynamic range $\sim$1.14$\times$ below that of upstream PNs, as expected (Prisco et al., 2021). Meanwhile, classification remains robust, with accuracy ranging from 46–78% across c = 0.3–5.0 (a 17-fold physical range), well above the 3.6% chance level and consistent with the behavioral concentration tolerance observed by Asahina et al. (2009). The prediction that KC population codes are concentration-invariant (Stopfer et al., 2003) receives some support, with cross-concentration centroid accuracy degrading only at very low concentrations, likely due to single-concentration training and the absence of ORN adaptation.

Sub-Additive KC Responses to Odor Mixtures: To test generalization beyond the single odors used for training, we presented 2–3 odor mixtures (element-wise sums of component OR patterns, 351 pairs—all combinations excluding the spontaneous firing rate—and 50 triplets per seed) to the trained models. For 2-odor mixtures, 69.4% of individual KCs respond less to the mixture than predicted by summing their responses to each component alone (sub-additive), closely matching the 73% reported for binary blends by Honegger et al. (2011); for 3-odor mixtures, 72.6% are sub-additive. Mixture KC representations are geometrically distinct from components (cosine similarity 0.806) and linearly separable by a support vector machine (73.9%), despite no mixture-specific training.

Connectome Dominates Learned Solutions

If the connectome genuinely constrains the model, independently trained models should converge to similar solutions. We find consistency in KC spike patterns, with mean pairwise cosine similarity of the trial-averaged KC patterns across models reaching 0.942. Weighted-average correlation across all parameters increases from teacher–teacher r = 0.781 to student–student r = 0.909 (Δ = +0.128), with the largest gains in KC thresholds (0.528 $\rightarrow$ 0.953) and decoder weights (0.788 $\rightarrow$ 0.903). Scalar parameters (synaptic strengths, gap conductances, g_{soma}, τ_{APL}) converge to CV 1.8% across students, down from 5.4% in the teacher. The connectome already constrains the rate-based teachers, but spiking dynamics impose additional convergence, demonstrating that both structure and biophysics jointly determine the solution.

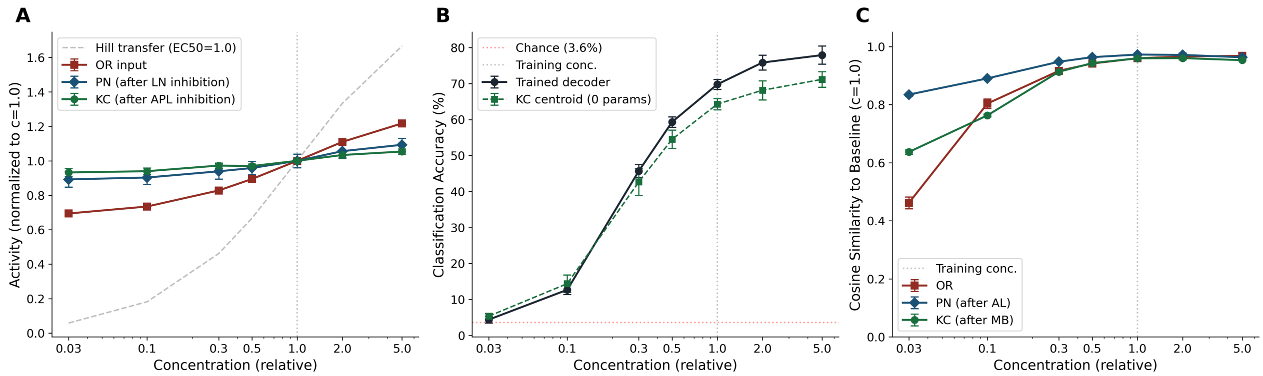

Figure 3: Concentration invariance and sparse KC coding (mean $\pm$ 1 s.d. plotted). (A) Normalized population activity across OR, PN, and KC layers as a function of odor concentration, showing progressive gain compression through successive circuit stages. The dashed line shows the Hill transfer function (EC_{50} = 1.0). (B) Odor classification accuracy versus concentration for the trained decoder and parameter-free KC centroid classifier. (C) Cosine similarity of layer-wise representations to the training concentration. PN representations maintain higher similarity to the training concentration than KCs do, although OR, PN, and KC representations all maintain relative stability across concentrations; the greater KC sensitivity reflects sparse coding selectivity, not a circuit failure.

Parameter Boundaries

All 366 spiking neurons are clamped to biological parameter ranges and most converge to the interior of the range. The main exception is KC spike thresholds, where $\sim$47% (66–69/144) hit the upper bound (-30 mV), driven by the two-compartment g_{soma} bottleneck and absence of KC gap junctions (Fig. S4A). This is KC-specific, as LNs favor the lower bound ($\sim$29%) and PNs stay interior ($\sim$83%). The fraction of KCs reaching this bound scales with task difficulty (11.9% at 7 odors, 35.3% at 14, 46.7% at 28, 55.8% at 56) (Fig. S4B), suggesting extreme KC thresholds are a sparsity mechanism. Gap junction and KC soma conductances are consistent across models (LN–LN 0.136 $\pm$ 0.003 nS, PN–PN 0.096 $\pm$ 0.001 nS, LN_e–PN 0.082 $\pm$ 0.002 nS, g_{soma} = 14.6 $\pm$ 0.1 nS).

Perturbation Analysis

Ablation Studies: To isolate individual circuit contributions, we performed three ablation experiments, each tested both prior to and after training. (i) Gap junctions were dispensable both when the model was retrained without them and when they were removed post-hoc, so our method for predicting electrical synapses absent from the connectome did not reveal a computational role. (ii) APL disabled: AL decorrelation tripled (15.2% vs 4.4%) while MB decorrelation dropped (31.5% vs 35.8%), shifting decorrelation upstream (Fig. 4); post-hoc removal doubled KC sparsity (29.4% vs 12.5%). (iii) STD removed: eliminated gain control (PN 2.75$\times$, KC 2.77$\times$ vs canonical 1.23$\times$, 1.14$\times$); post-hoc removal dropped accuracy and doubled KC sparsity.

Shuffled Connectome: To investigate the role of the connectome wiring beyond degree distribution, we shuffled the connectome by preserving per-neuron-type connection counts but randomizing specific synaptic partners, then retrained and analyzed the model. As seen in Fig. 4, AL and MB decorrelation became highly variable (4.3 $\pm$ 10.7% and 30.3 $\pm$ 5.4%, respectively), showing specific wiring carries functional information beyond degree distribution. This conclusion is further reinforced by a reduction in APL-mediated KC suppression (33.4 $\pm$ 10.9%) and sub-additive mixture coding KC responses (43.8 $\pm$ 7.0%) within the shuffled connectome.

Sparsity Loss for Successful Training: Our training included KC sparsity as a constraint, in addition to the task-motivated cross-entropy loss for odor classification. The sparsity loss might be considered to approximate metabolic energy constraints absent from the model, but it is possible that KC sparsity could emerge from connectivity constraints, e.g., the broad inhibitory feedback from the APL. To investigate further, we tested three constraint forms in the loss function: (i) no sparsity loss, (ii) an all-neuron energy penalty, and (iii) a KC-specific energy penalty. (i) No sparsity loss: the optimizer drives 80% KC activity and eliminates MB decorrelation, overwhelming APL inhibition. (ii) All-neuron energy penalty: fails because ORNs and LNs have fewer downstream synapses than KCs, so suppressing them costs less gradient and the optimizer silences the AL rather than sparsifying KCs. (iii) KC-specific energy penalty: preserves AL processing while producing biological sparsity and decorrelation. We therefore assume the KC-specific loss proxies for cell-type-specific homeostatic dynamics and regulatory mechanisms not captured by the connectome.

KC-specific sparsity enforced by the loss function underlies the emergence of decorrelation, while the perturbations in Fig. 4 demonstrate that the APL and the specific connectome-constrained wiring keep that decorrelation localized to the MB, as removing the APL drives decorrelation upstream into the AL, and shuffling the connectome gives highly variable decorrelation.

Discussion

Connectome-derived rate-based models of the adult *Drosophila* visual system have previously been trained and validated against existing biological data (Lappalainen et al., 2024). Our work extends this approach while including several features not previously combined in a single connectome-constrained model. We use spiking LIF neurons throughout the AL alongside a graded (non-spiking) APL interneuron, modeling two biophysically distinct neuron types within this circuit. KCs are modeled with two compartments (dendritic and axonal) coupled by a learnable soma conductance, capturing the anatomical separation of the MB calyx and lobes. Electrical gap junctions with proxy topologies derived from the connectome are incorporated, allowing the model to learn gap junction conductances alongside chemical synaptic strengths. Also included are Tsodyks–Markram short-term depression on all chemical synapses (Tsodyks & Markram, 1997), excitatory and inhibitory LN subtypes (Berck et al., 2016), and KC–KC lateral axo-axonic excitation (Winding et al., 2023). Finally, we train the model using an ANN-to-SNN transfer learning pipeline, first training a rate-based model with standard backpropagation, then fine-tuning parameters with surrogate gradient descent.

After learning 449 biophysical parameters within biological bounds, this structurally faithful connectome-constrained model spontaneously reproduces multiple independently measured circuit properties, including decorrelation localized to the MB, calibrated APL inhibition, sub-additive coding of odor mixtures, and a degree of concentration invariance. None of these properties were part of the training objective, meaning they emerged from the interaction of connectome structure, biophysical constraints, and a simple classification task. This convergence to biologically realistic solutions suggests that the connectome imposes some functional constraints that narrow the space of viable solutions. Further, connectome-constrained optimization offers a principled framework that leverages large-scale anatomical data to infer unmeasured biophysical parameters such as g_{soma}, gap junction conductances, and synaptic depression time constants from task performance alone, yielding testable predictions that can guide future experiments.

Beyond reproducing known biology, this methodology provides a trained model that serves as a platform for *in silico* experiments able to generate testable hypotheses. Our perturbation studies imply that the APL localizes decorrelation to the MB, that short-term depression plays a role in concentration-dependent gain control, and that specific topology encodes function beyond degree distribution. These constitute predictions that could be tested by targeted optogenetic or pharmacological manipulation in the larval circuit.

Limitations and Future Work

There are important caveats to interpreting the resulting models. The approach requires a meaningful training objective, and ours (simple odor classification) is a proxy for the larva's actual behavioral objectives, which likely encompass a richer set of demands that would require downstream circuitry (mushroom body output neurons, motor neurons) not included in the current model. Further, connectome-constrained optimization cannot prove that the biological circuit uses the same parameter values our models learn. It can only suggest that the connectome, with task and parameter constraints, permits biologically realistic solutions within those ranges. It is also unable to capture properties absent from the connectome data, including neuromodulation, experience-dependent structural plasticity, or synaptic connections below the detection threshold of EM reconstruction.

The hyperparameter tuning required for stable spiking network training—learning rates, noise levels, gradient clipping—is itself not biologically derived, representing an engineering necessity. Similarly, the use of surrogate gradient training and transfer learning is a methodological choice, not a biological claim. Gradient-based optimization naturally drives increased spiking activity to maximize classification information, opposing the biological sparsity maintained by APL inhibition. The explicit KC-specific sparsity loss is needed to counteract this pressure and serves as a proxy for cell-type-specific homeostatic mechanisms absent from the model.

There are nevertheless several directions to further explore with the existing model. The olfactory circuit can be extended to include additional neurons in the MB, specifically dopaminergic neurons and mushroom body output neurons involved in learned responses to odors. There is a substantial body of neural imaging data for this circuit that could be used for validation. Also of interest would be to extend the pathway from the AL via the MB to descending neurons controlling motor behavior and directly compare circuit outputs to behavioral data such as steering responses in odor gradients. For larger connectomes, a piecewise training strategy (training individual subcircuits independently before end-to-end fine-tuning) may manage the increased parameter count and gradient flow. Connectome-derived spiking circuits could also

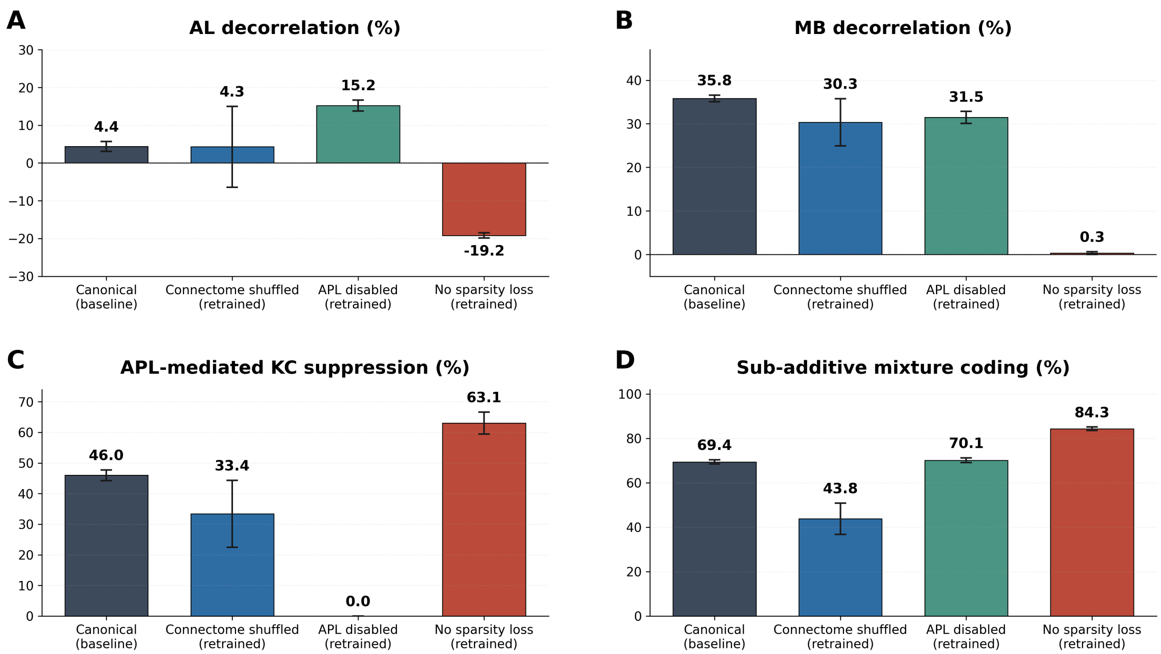

Figure 4: Circuit-component perturbations clarify each architectural feature's contribution, comparing the canonical model against three retrained conditions: shuffled connectome, APL-disabled, and no-sparsity-loss (five-seed mean $\pm$ 1 s.d.). (A) AL decorrelation rises sharply when the APL is removed, indicating that the APL maintains MB-localized decorrelation. Shuffling the connectome instead makes AL decorrelation highly variable—demonstrating that specific topology beyond degree distribution stabilizes and minimizes AL decorrelation—while removing the sparsity loss drives it strongly negative. (B) MB decorrelation is mostly preserved (although still less localized overall) under APL ablation. The shuffled connectome is again less stable than the canonical model for MB decorrelation, which collapses entirely without the sparsity loss. (C) The APL-mediated KC suppression of $\sim$50% reported by Mancini et al. (2023) falls to 0% when the APL is removed (confirming the ablation), is reduced and destabilized by shuffling the connectome, and is inflated in the densely ($\sim$80%) active KC regime without the sparsity loss. (D) Sub-additive mixture coding of $\sim$73% is reported by Honegger et al. (2011). The fraction of KCs whose 2-odor mixture response falls below the sum of their single-odor responses is unchanged by APL ablation, indicating it arises from other circuit components. It drops and is destabilized when the connectome is shuffled (again demonstrating the importance of connectome topology) and rises in the dense no-sparsity regime.

serve as biologically grounded priors for energy-efficient, low-latency, event-driven neuromorphic systems (Davies et al., 2021; Roy et al., 2019; Zhang et al., 2023).

Conclusion

This work demonstrates that connectome-constrained spiking neural networks can reproduce emergent computations in the larval *Drosophila* olfactory pathway. The Winding et al. (2023) connectome combined with biophysical constraints and task-driven optimization gives rise to decorrelation, appropriate suppression of KCs by the APL, sub-additive mixture coding, and concentration-invariant gain control that each match independent biological measurements. Our ensemble of five independently trained models converges to consistent solutions, demonstrating the role of the connectome in determining circuit function.

The methodology developed here (connectome-fixed topology, minimal learnable parameters, biophysical clamping, and independent validation) is not specific to the larval *Drosophila* olfactory system. As connectomes become available for larger circuits, such as the entire adult *Drosophila* brain (Dorkenwald et al., 2024; Schlegel et al., 2024) and the mouse visual cortex (MICrONS-Consortium, 2025), scaling will become a critical question as we determine which biological measurements and task-based training methods can constrain the much larger parameter space. The component-perturbation framework introduced here is applicable to other connectome-constrained models and provides a principled approach to generating experimentally testable hypotheses from anatomical data.

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Supplementary Materials

Table S1: Odor Index Reference

ID	Odor name	ID	Odor name
1	spontaneous firing rate	15	4-methylphenol
2	propanoic acid	16	1-butanol
3	geranyl acetate	17	1-hexanol
4	(1R)-(-)-fenchone	18	1-octen-3-ol
5	E2-hexenal	19	1-heptanol
6	2-heptanone	20	3-octanol
7	2,3-butanedione	21	1-nonanol
8	cyclohexanone	22	ethyl acetate
9	methyl salicylate	23	propyl acetate
10	benzaldehyde	24	pentyl acetate
11	acetophenone	25	isopentyl acetate
12	2-methylphenol	26	ethyl butyrate
13	methoxybenzene	27	octyl acetate
14	4-allyl-1,2-dimethoxybenzene	28	carbon dioxide

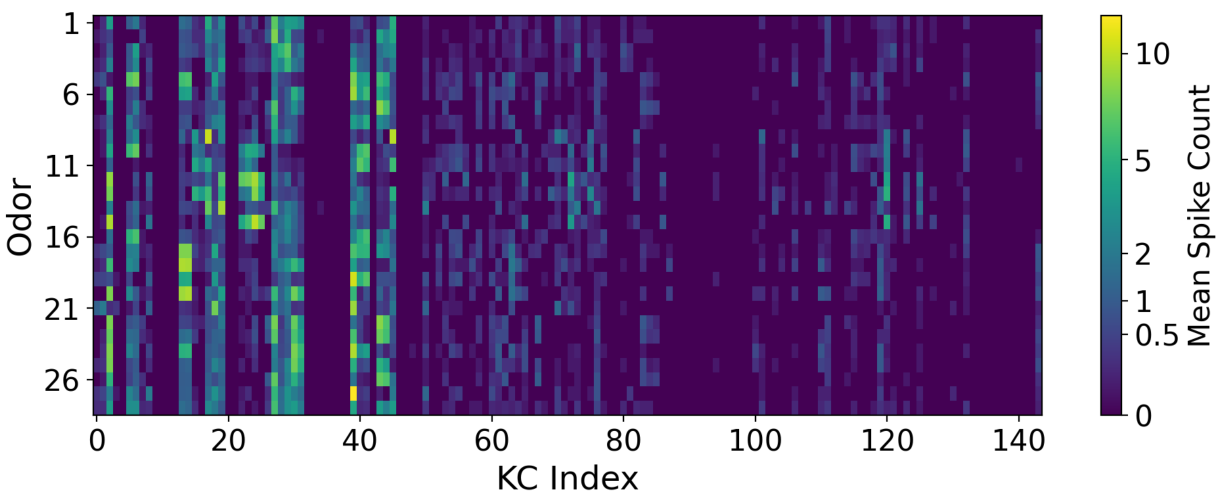

Figure S1: The mean KC activity heatmap shows how a sparsely active neural population can encode odorant representations, with certain KCs more responsive to aromatic compounds (e.g. KCs 22–25) and others to aliphatic compounds (e.g. KCs 28–31).

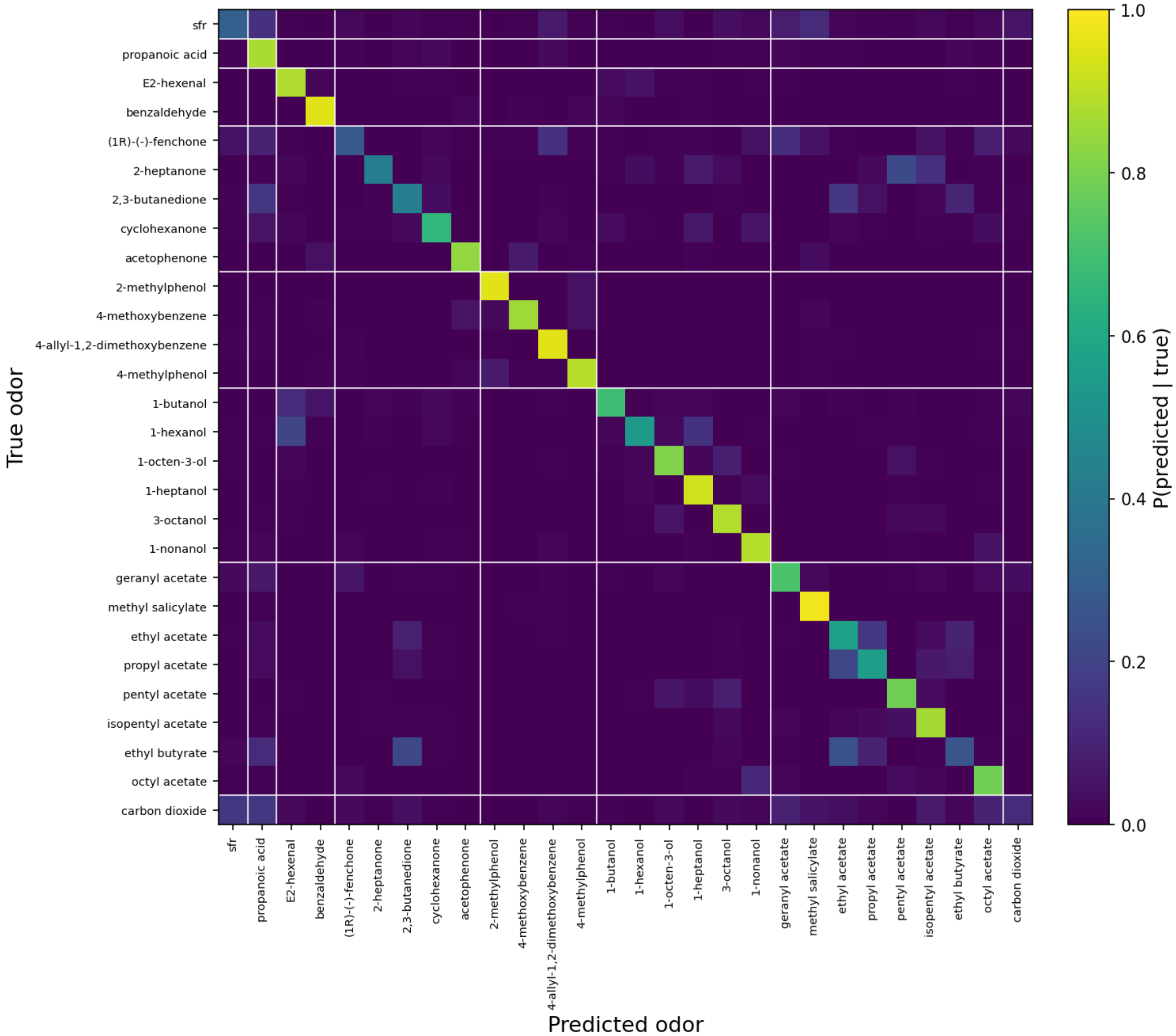

Figure S2: Mean classification confusion matrix (five models, 50 noisy trials per odor). Rows are the true odor, columns the predicted odor, and cell color is $P(\text{predicted} \mid \text{true})$. Odors are grouped by chemical family, with white lines marking family boundaries. The strong diagonal (mean ≈ 0.70) reflects the overall classification accuracy, while off-diagonal errors concentrate *within* chemical families—most visibly among the esters and alcohols. The no-odor baseline (sfr) and CO_2, which lacks a dedicated receptor among the modeled ORs, are the diffuse, weakly classified rows.

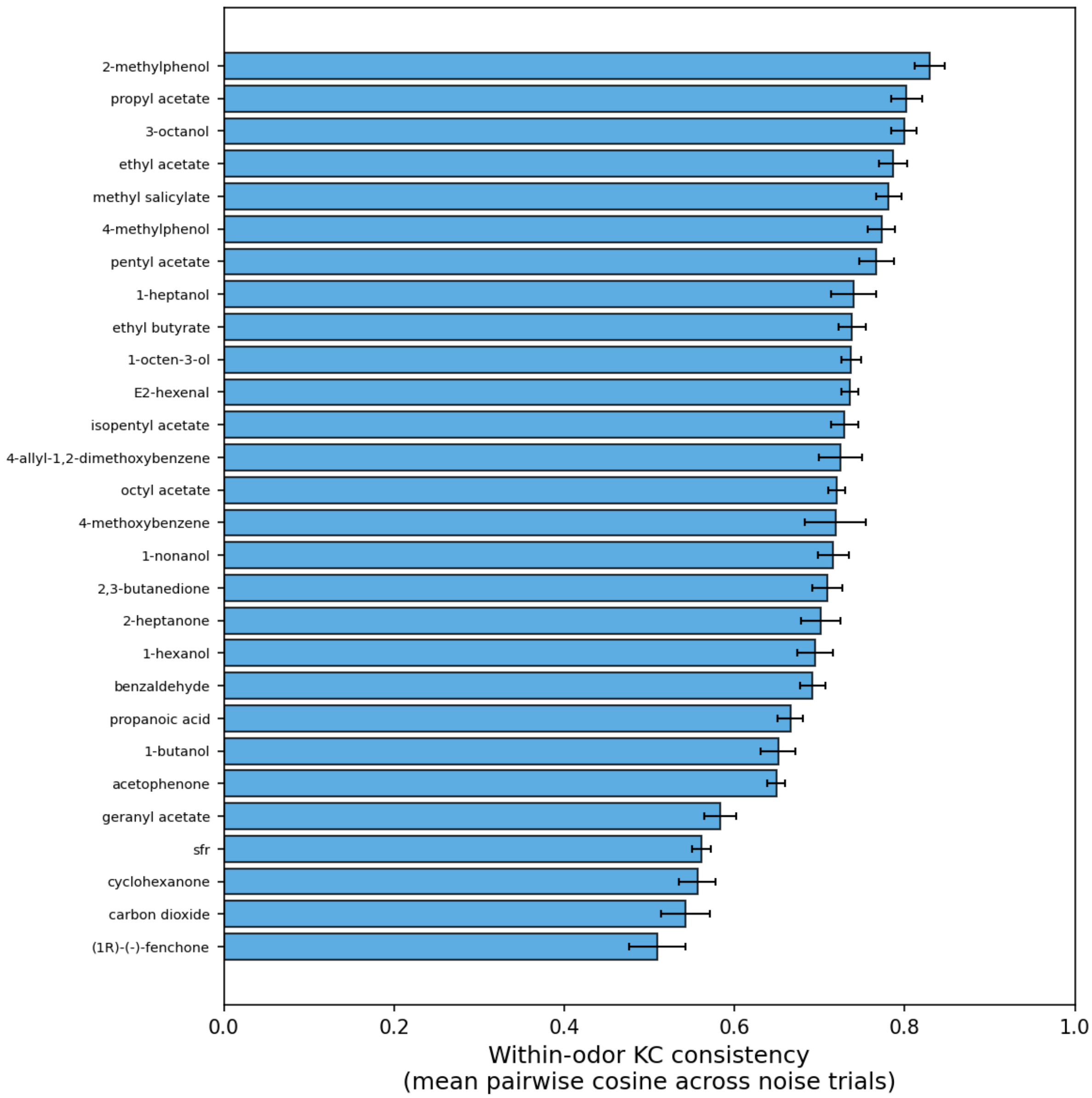

Figure S3: Within-odor KC representational consistency across noise realizations. For each odor, single-trial KC spike-count vectors from 50 noisy presentations are compared by mean pairwise cosine similarity and averaged across the five models, giving a grand mean of 0.701 across all odors. Consistency is lowest for weak-response or ambiguous odors (e.g. (1R)-(-)-fenchone, CO_2, and the sfr baseline).

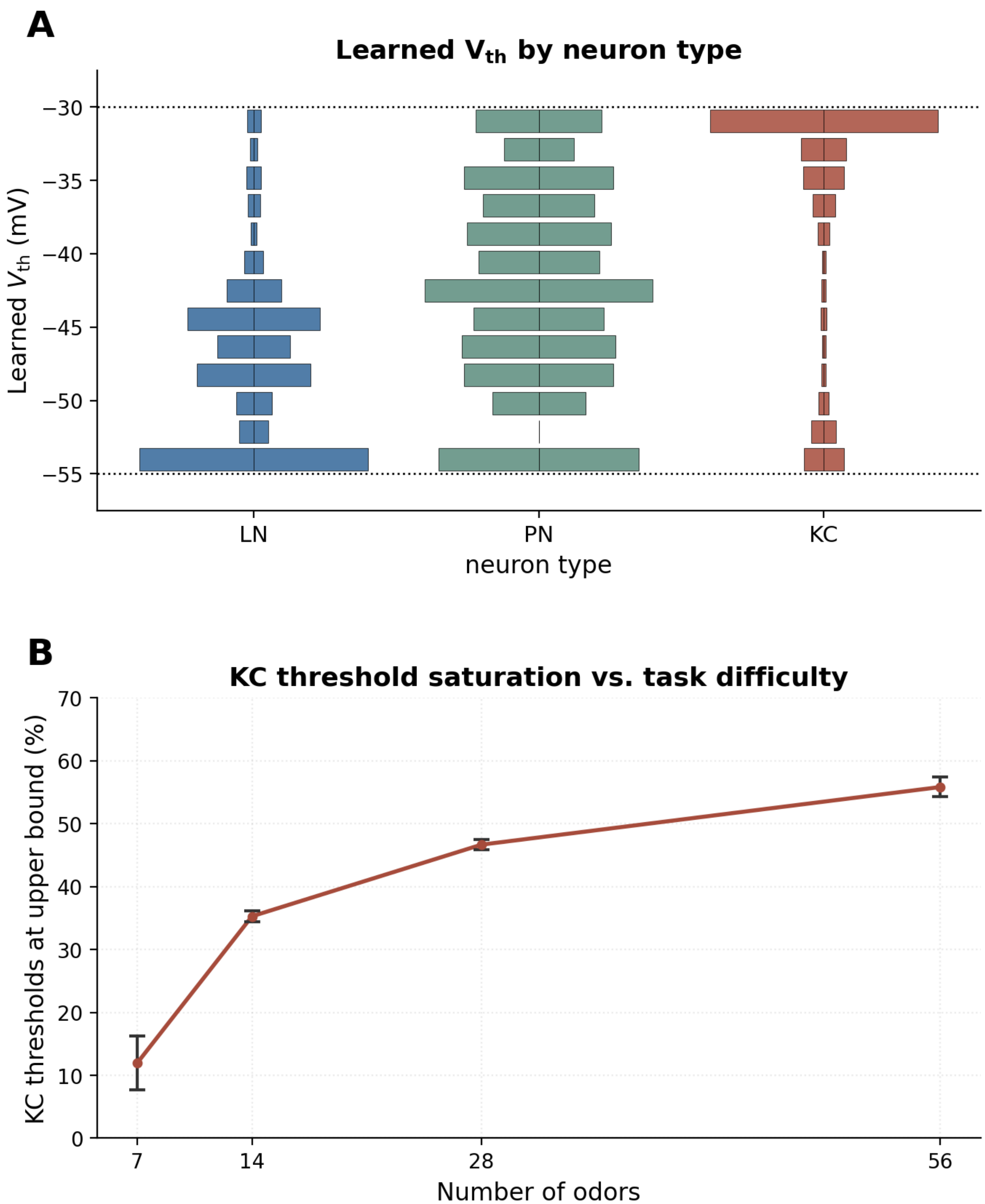

Figure S4: KC firing-threshold adaptation. (A) Fraction of Kenyon-cell thresholds pinned at the -30 mV upper clamp bound as a function of the number of odors the network must classify (7, 14, 28, 56; mean $\pm$ s.d. across the five seeds). Saturation grows with task difficulty: harder discrimination pushes a larger fraction of KCs toward maximal excitability. (B) Distributions of the learned membrane threshold V_{th} by neuron population (LN, PN, KC), pooled across the five models. Dotted lines mark the $[-55, -30]$ mV clamp. The populations occupy distinct, biologically plausible bands well inside the bounds.